

Size:104x152mm

IDAMAN PHARMA MANUFACTURING

AMILLIN CAPSULES 250 MG

COMPOSITION:

Ampicillin Trihydrate equivalent to 250 mg of Ampicillin

DESCRIPTION

Clean, uniform, dry telescope type hard gelatin capsules. Must be locked, shiny & not smudged or sticky. Must be firm & fully packed. Printed "Amillin 250" & "PM" on the capsules.
Capsule: Cap : Opaque Black
Body : Opaque Scarlet
Size : No. 2
Powder: A white coarse powder

PHARMACODYNAMICS

Bactericidal for Gram-positive organisms, acts by inhibiting transpeptidase, the enzyme responsible for cross-linking of peptidoglycan during the final stage of synthesis of the bacterial cell wall, thus exerts its effect against actively dividing bacteria. Bacteriostatic for Gram-negative organisms; action resembles those of chloramphenicol and tetracyclines, i.e. by inhibition of bacterial protein synthesis.

Ampicillin resembles benzylpenicillin in its action against Gram-positive organisms, including *Streptococcus pneumoniae* and other streptococci, but with the possible exception of activity against *Enterococcus faecalis*, it is slightly less potent than benzylpenicillin. *Listeria monocytogenes* is highly sensitive.

The Gram-negative cocci *Moraxella catarrhalis* (*Branhamella catarrhalis*), *Neisseria gonorrhoeae* and *Neisseria meningitidis* are sensitive.

Ampicillin is more active than benzylpenicillin against some Gram-negative bacilli, including *Haemophilus influenzae* and *Enterobacteriaceae* such as *Escherichia coli*, *Proteus mirabilis*, *Salmonella* and *Shigella* spp. It is inactive against *Pseudomonas aeruginosa*.

Ampicillin also has activity similar to benzylpenicillin against other organisms including many anaerobes and *Actinomyces* spp.

There is synergy against some beta-lactamase-producing organisms between ampicillin and beta-lactamase inhibitors and also penicillinase-stable drug. Synergy has also been shown between ampicillin and aminoglycosides against a range of organisms, including enterococci. Variable effects ranging from synergy antagonism have been reported between ampicillin and other beta lactams, bacteriostatic drugs such as chloramphenicol and rifampicin.

Like benzylpenicillin, ampicillin is inactivated by beta lactamases, although other mechanisms may be responsible for resistance in some species. There are geographical variations in the incidence of resistance, but most staphylococci and many strains of *E. coli*, *H. influenzae*, *M. catarrhalis*, *N. gonorrhoeae*, and *Salmonella* and *Shigella* spp. are resistant.

PHARMACOKINETICS

Ampicillin is acid stable and 30 to 60% of an oral dosed is absorbed. Peak serum concentration of 0.8 to 4 mcg/ml are attained 1 to 2 hours after oral doses of 125 to 500 mg of anhydrite or trihydrate. The serum half-life is 1 to 2 hours which is increased in subjects with renal function impairment. Ampicillin enters most body cavities, tissues and fluids except the cerebrospinal fluid unless the meninges are inflamed, in which case large amounts may be detectable in this fluid. Ampicillin crosses the placenta and is excreted in the milk; volume of distribution about 20 L which in subjects with cirrhosis may reach about 60 L. About 20% of Ampicillin are bound to plasma proteins. In normal subjects approximately 40% (ampicillin) of the dose administered is metabolised. 30% to 50% of an oral dose is excreted in the urine in 6 to 8 hours; up to 70% is excreted in 24 hours; urinary excretion is decreased after the administration of probenecid. Ampicillin is excreted in the bile in concentrations up to 4 mcg/ml after an oral dose of 125 mg; biliary excretion is increased in subject with cirrhosis.

INDICATIONS

Ampicillin is a broad-spectrum penicillin, indicated for the treatment of a wide range of bacterial infections caused by ampicillin sensitive organisms. Typical indications include ear, nose and throat infections, bronchitis, pneumonia, urinary tract infections, gonorrhoea, gynaecological infections, septicaemia, peritonitis, endocarditis, meningitis and enteric fever and gastro-intestinal infections. Parenteral usage is indicated where oral dosage is inappropriate.

DOSAGE AND ADMINISTRATION

Infections:

i) Ear, nose, throat infections
Adults and elderly patients: 250 mg four times a day.

ii) Bronchitis
Adults and elderly patients: routine therapy 250 mg four times a day.

iii) Pneumonia
Adults and elderly patients: 500 mg four times a day.

iv) Urinary tract infections
Adults and elderly patients: 500 mg three times a day.

v) Gastrointestinal infections
Adults and elderly patients: 500 to 750 mg three to four times daily.

vi) Enteric infections
Adults and elderly patients: acute (1-2 g four times a day for two weeks); carries (1-2 g four times a day for four to twelve weeks).

vii) Gonorrhoea
Adults and elderly patients: 2 g orally with 1 g of probenecid as a single dose. Repeated doses are recommended for the treatment of females.

Paediatric population

Children less than 10 years old may use half adult routine dosage. All recommended dosages are a guide only. In severe infections, the above dosages may be increased, or ampicillin given by injection.

Renal Impairment

In the presence of severe renal impairment (creatinine clearance less than 10 ml/min) a reduction in dose or extension of dose interval should be considered. In cases of dialysis, an additional dose should be administered after the procedure.

ROUTE OF ADMINISTRATION

Oral

CONTRAINDICATIONS

Hypersensitivity to the active substance, beta-lactam antibiotics.

WARNINGS AND PRECAUTIONS

Hypersensitivity

Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients receiving therapy with beta-lactams. Before initiating therapy with ampicillin, careful inquiry should be made concerning previous hypersensitivity reaction to penicillins, cephalosporins, carbapenems or other beta-lactam agents. If an allergic reaction occurs, ampicillin must be discontinued immediately and appropriate alternative therapy instituted.

Serious anaphylactoid reactions require immediate emergency treatment with epinephrine, oxygen, intravenous steroids and airway management, including intubation, should also be administered as indicated.

Infectious mononucleosis/ acute or chronic leukaemia

Ampicillin should be avoided if infectious mononucleosis and/or acute or chronic leukaemia of lymphoid origin are suspected. The occurrence of a skin rash has been associated with these conditions following the administration of ampicillin.

Gastrointestinal disease

Ampicillin should also be used in cautious with patients that have history of gastrointestinal disease, especially ulcerative colitis, regional enteritis or antibiotic-associated colitis. Pseudomembranous colitis has been reported with the use of antibiotics and may range in severity from mild to life-threatening. Therefore, it is important to consider its diagnosis in patients who develop diarrhoea during or after antibiotic use. If prolonged or significant diarrhoea occurs or the patient experiences abdominal cramps, treatment should be discontinued immediately and the patient investigated further.

Renal impairment

The dosage of ampicillin will need to be reduces in patients with marked renal impairment.

Acute interstitial nephritis

Ampicillin should be avoided in patients who have experienced acute interstitial nephritis as a result of antibiotic therapy.

Colonisation

The use of ampicillin can lead to the development of severe colitis as a result of colonisation with a toxin-producing organism. The colitis can be fatal and must be treated promptly. If significant diarrhoea occurs, ampicillin should be discontinued. This may be sufficient treatment in the early stages although cholestyramine orally may help by binding the toxin to the colonic lumen. In severe cases oral vancomycin 250 mg 6 hourly for 5-10 days has proved effective. Parenteral vancomycin is not effective.

Allergic diathesis

Extreme caution should be used in the treatment of patients within allergic diathesis.

Suprainfection

Suprainfection with other mycotic organisms or bacteria may occur during therapy and if it occurs, discontinue ampicillin and substitute with appropriate treatment.

Prolonged and high-dose therapy

Prolonged use may occasionally result in overgrowth of non-susceptible organisms. As with any potent agent, periodic assessment of organ function should be made during prolonged therapy. Care is also necessary if very high doses of penicillins are given, especially if renal function is poor, because of the risk of neurotoxicity. Renal, hepatic and hematological status should be monitored during prolonged and high-dose therapy. In renal impairment the safe dosage may be lower than usually recommended.

INTERACTION WITH OTHERS MEDICAMENTS

Concurrent administration of allopurinol during treatment with ampicillin can increase the likelihood of allergic skin reactions.

Probenecid decreases the renal tubular secretion of ampicillin. Concurrent use with ampicillin may result in decreased and prolonged blood levels of ampicillin.

Bacteriostatic drugs may interfere with the bactericidal action of ampicillin.

In common with other oral broad-spectrum antibiotics, ampicillin may reduce the efficacy of oral contraceptives and patients should be warned accordingly.

It is recommended that when testing for the presence of glucose in urine during ampicillin treatment, enzymatic glucose oxidase methods should be used. Due to the high urinary concentrations of ampicillin, false positive readings are common with chemical methods.

PREGNANCY AND LACTATION

Pregnancy

Animal studies with Ampicillin have shown no teratogenic effects. The product has been in extensive clinical use and its use in human pregnancy has been well documented in clinical studies. When antibiotic therapy is required during pregnancy, Ampicillin may be considered appropriate.

Lactation

Ampicillin is excreted in trace amounts in human milk. Therefore, caution should be exercised when ampicillin-class antibiotics are administered to a nursing woman.

ADVERSE EFFECTS

Hypersensitivity

Hypersensitivity reactions are the most common adverse effects and if it occurs, the treatment should be discontinued. Skin rash, pruritis, urticaria, maculopapular (often appear about 5 days after treatment has finished), itching and wheezing have been reported occasionally. The incidence is higher in patients suffering from infectious mononucleosis and acute or chronic leukaemia of lymphoid origin. Purpura has also been reported. Rarely, skin reactions such as erythema multiforme and Stevens-Johnson syndrome and toxic epidermal necrolysis have been reported. As with other antibiotics, anaphylaxis has been reported rarely. Serious and occasionally fatal anaphylactoid reactions have been reported in patients receiving therapy with beta-lactams. It is more frequent following parenteral therapy but may also occur with oral therapy.

Renal effects

Interstitial nephritis can occur rarely.

Gastrointestinal reactions

Effects include abdominal or stomach cramps, pain, bloating, nausea, vomiting and diarrhoea. Pseudomembraneous colitis and haemorrhagic colitis have been reported rarely.

Haematological effects

As with other beta-lactams, haematological effects including transient leucopenia, transient thrombocytopenia and haemolytic anaemia have been reported rarely. Prolongation of bleeding time and prothrombin have also been reported rarely.

Suprainfection

Suprainfection with Pseudomonas and Candida with prolonged use.

Nervous system disorders

Myoclonus and convulsions.

Others

Fever, increased thirst, unusual tiredness or weakness.

In common with other antibiotics, Ampicillin may affect the gut flora, leading to lower oestrogen reabsorption and reduced efficacy of combined oral contraceptives.

OVERDOSE AND TREATMENT

i) Overdose may cause gastrointestinal tract (GIT) disturbances such as nausea, vomiting and diarrhoea as well as reversible impairment of liver function. In cases of overdose, discontinue medication, treated symptomatically and institute supportive measures as required. Treatment includes stopping antibiotic treatment and treating GIT distress with milk every 3 hours alternated with Bismuth Subcarbonate 5 g every 3 hours.

ii) Gross overdose will produce very high urinary concentration. Problems are unlikely if adequate fluid intake and urinary output are maintained; however, crystalluria is a possibility. More specific measures may be necessary in patients with impaired renal function; the antibiotic is removed by haemodialysis but not peritoneal dialysis.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

Adverse effects on the ability to drive or operate machinery have not been observed.

STORAGE CONDITIONS

Keep container tightly closed in a dry place, below 30°C. Protect from light.

SHELF LIFE

3 years from the date of manufacture.

PRESENTATION

Packed in blister of 10 capsules in carton box (50 x 10's)

REGISTRATION NUMBER

MAL19860005AZ

For further information, please consult your doctor or pharmacist.

Date Revision: 7th August 2020

PP-LT097.05 Product Registration Holder/ Manufacturer;

IDAMAN PHARMA MANUFACTURING SDN BHD (200401023395)
LOT 120, TAMAN FARMASEUTIKAL,
32610 BANDAR SERI ISKANDAR,
PERAK DARUL RIDZUAN,
MALAYSIA.